

Approximation of Bandlimited and Almost-Bandlimited Functions Using PSWFs

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1 Introduction

Prolate spheroidal wave functions (PSWFs) were introduced in the classical work of Slepian, Pollak, and Landau in connection with time and frequency concentration [1, 2]. They are especially useful when a function is bandlimited and we are interested in its behavior on a finite interval.

In this short note we use one-dimensional PSWFs to approximate two nontrivial functions on $[-1, 1]$. The first function is exactly bandlimited. The second is not exactly bandlimited, but most of its frequency content is concentrated in the same frequency range.

2 PSWFs and the Time–Band Limiting Operator

We use the Fourier transform convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

Let P_c denote the projection onto frequencies in $[-c, c]$, and let Q denote restriction to $[-1, 1]$. The operator QP_c is

$$(QP_c f)(x) = \int_{-1}^1 \frac{\sin(2\pi c(x-t))}{\pi(x-t)} f(t) dt.$$

At $x = t$, the kernel is defined by continuity:

$$\lim_{t \rightarrow x} \frac{\sin(2\pi c(x-t))}{\pi(x-t)} = 2c.$$

The PSWFs ψ_n^c satisfy

$$QP_c \psi_n^c = \lambda_n^c \psi_n^c.$$

This connection with the time–band limiting operator is one of the main reasons PSWFs are useful for approximation of bandlimited functions [3].

3 Finite PSWF Approximation

Assume that the PSWFs are normalized in $L^2([-1, 1])$. Define

$$V_N^c = \text{span}\{\psi_0^c, \psi_1^c, \dots, \psi_{N-1}^c\}.$$

The orthogonal projection onto this space is denoted by Π_N^c :

$$\Pi_N^c f = \sum_{n=0}^{N-1} \langle f, \psi_n^c \rangle \psi_n^c.$$

Therefore the N -term PSWF approximation is

$$f_N(x) = \Pi_N^c f(x) = \sum_{n=0}^{N-1} a_n \psi_n^c(x),$$

where

$$a_n = \langle f, \psi_n^c \rangle = \int_{-1}^1 f(x) \overline{\psi_n^c(x)} dx.$$

For the real-valued examples used here,

$$a_n = \int_{-1}^1 f(x) \psi_n^c(x) dx.$$

Because the PSWFs are orthonormal,

$$\|f - f_N\|_{L^2(-1,1)}^2 = \sum_{n=N}^{\infty} |a_n|^2.$$

We report both the absolute L^2 error

$$E_N = \|f - f_N\|_{L^2(-1,1)}$$

and the relative L^2 error

$$E_N^{\text{rel}} = \frac{\|f - f_N\|_{L^2(-1,1)}}{\|f\|_{L^2(-1,1)}}.$$

How the PSWFs are computed

The MATLAB code uses the matrix produced by `prolatematrix.m`. This matrix represents the PSWF differential operator in a normalized Legendre basis.

Let P_k be the usual Legendre polynomial of degree k . We use

$$\phi_k(x) = \sqrt{k + \frac{1}{2}} P_k(x).$$

These functions satisfy

$$\int_{-1}^1 \phi_j(x) \phi_k(x) dx = \delta_{jk}.$$

If

$$v_n = (v_{0n}, v_{1n}, \dots, v_{m-1,n})^T$$

is an eigenvector of the matrix returned by `prolatematrix.m`, then the corresponding PSWF is approximated by

$$\psi_n^c(x) \approx \sum_{k=0}^{m-1} v_{kn} \phi_k(x).$$

Thus the Legendre polynomials are only used to build the PSWFs numerically. The final approximation of the target function is still a PSWF expansion.

How the coefficients are computed

The exact coefficient is

$$a_n = \int_{-1}^1 f(x) \psi_n^c(x) dx.$$

In MATLAB we evaluate this integral with Gauss–Legendre quadrature. If x_j and w_j are the quadrature nodes and weights, then

$$a_n \approx \sum_{j=1}^{N_q} w_j f(x_j) \psi_n^c(x_j).$$

We use $N_q = 700$ quadrature points. In matrix form, MATLAB computes all coefficients at once using

$$\mathbf{a} = \mathbf{PsiQ}' * (\mathbf{wq} * \mathbf{fQ}).$$

Gauss–Legendre quadrature is a natural choice here because all integrals are over $[-1, 1]$, and the PSWFs themselves are represented through Legendre polynomials. An N_q -point Gauss–Legendre rule integrates every polynomial of degree at most $2N_q - 1$ exactly.

The same quadrature rule is used for the L^2 error:

$$\|f - f_N\|_2 \approx \left(\sum_{j=1}^{N_q} w_j |f(x_j) - f_N(x_j)|^2 \right)^{1/2}.$$

Therefore the code uses Gauss–Legendre quadrature both for the PSWF coefficients and for the final error calculation. The equally spaced grid is used only for plotting.

4 Which Functions Are Well Approximated by PSWFs?

PSWFs have a particularly strong connection with bandlimited functions. If

$$PW_c = \left\{ F \in L^2(\mathbb{R}) : \text{supp } \widehat{F} \subset [-c, c] \right\},$$

then the first N PSWFs form an optimal N -dimensional linear space, in a minimax sense, for the restrictions to $[-1, 1]$ of functions in the unit ball of PW_c . The corresponding Kolmogorov N -width is

$$d_N = \sqrt{\lambda_N^c}$$

[4].

For a single function, the practical condition is simpler: the tail of its PSWF coefficients should become small.

PSWFs can also work well for almost-bandlimited functions. Write

$$f = P_c f + (I - P_c) f.$$

Since

$$Qf = QP_c f + Q(I - P_c) f,$$

we have

$$Qf - \Pi_N^c Qf = (I - \Pi_N^c) QP_c f + (I - \Pi_N^c) Q(I - P_c) f.$$

Taking norms gives

$$\|Qf - \Pi_N^c Qf\|_2 \leq \|(I - \Pi_N^c) QP_c f\|_2 + \|(I - \Pi_N^c) Q(I - P_c) f\|_2.$$

Because Q and $I - \Pi_N^c$ are contractions,

$$\|(I - \Pi_N^c)Q(I - P_c)f\|_2 \leq \|(I - P_c)f\|_2.$$

For the bandlimited part $P_c f$, the PSWF singular-value structure gives

$$\|(I - \Pi_N^c)Q P_c f\|_2 \leq \sqrt{\lambda_N^c} \|P_c f\|_2.$$

Therefore

$$\|Qf - \Pi_N^c Qf\|_2 \leq \|(I - P_c)f\|_2 + \sqrt{\lambda_N^c} \|P_c f\|_2.$$

The first term measures the part of f outside the frequency band, while the second term is the PSWF truncation term. This explains why PSWFs can remain useful when a function is close to bandlimited, but not exactly bandlimited [5].

5 Example 1: An Exactly Bandlimited Function

We take $c = 10$ and consider

$$f_{\text{BL}}(x) = \sum_{j=1}^{12} A_j \operatorname{sinc}(2B_j(x - \tau_j)) \cos(2\pi\xi_j x + \phi_j),$$

where

$$\operatorname{sinc}(y) = \frac{\sin(\pi y)}{\pi y}.$$

Each shifted sinc term is bandlimited. Multiplication by the cosine shifts its Fourier support to frequencies centered at $\pm\xi_j$. The parameters in the MATLAB code satisfy

$$|\xi_j| + B_j < 10.$$

Therefore the whole function belongs to PW_{10} .

Using the first $N = 44$ PSWFs, the code computes

$$E_{44} = 3.237082e - 03, \quad E_{44}^{\text{rel}} = 2.276444e - 03.$$

6 Example 2: An Almost-Bandlimited Function

The second example is

$$f_{\text{ABL}}(x) = e^{-\rho x^2} \sum_{j=1}^{20} a_j \cos(2\pi\nu_j x + \phi_j) + \sum_{k=1}^5 b_k e^{-\gamma_k(x - \tau_k)^2}.$$

This function contains twenty oscillatory components and five localized Gaussian components. It is not exactly bandlimited because Gaussian Fourier transforms have nonzero tails at all frequencies. The parameters are chosen so that most of the frequency content lies near the band $[-10, 10]$.

Using the first $N = 44$ PSWFs, the code computes

$$E_{44} = 1.288079e - 02, \quad E_{44}^{\text{rel}} = 1.995680e - 02.$$

These two examples show the main idea of this short project. PSWFs give a natural finite-dimensional approximation space for exactly bandlimited functions, and they can also be effective for functions whose frequency content is strongly concentrated in the same band.

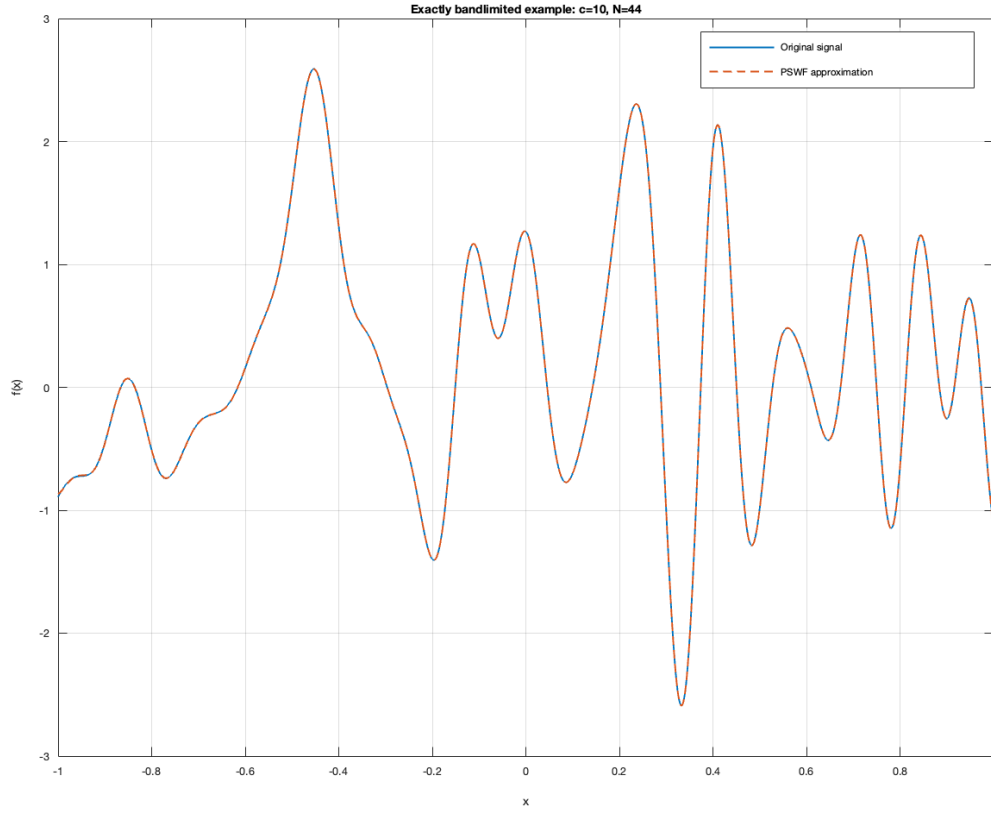


Figure 1: Exactly bandlimited test function and its approximation using the first 44 PSWFs with $c = 10$.

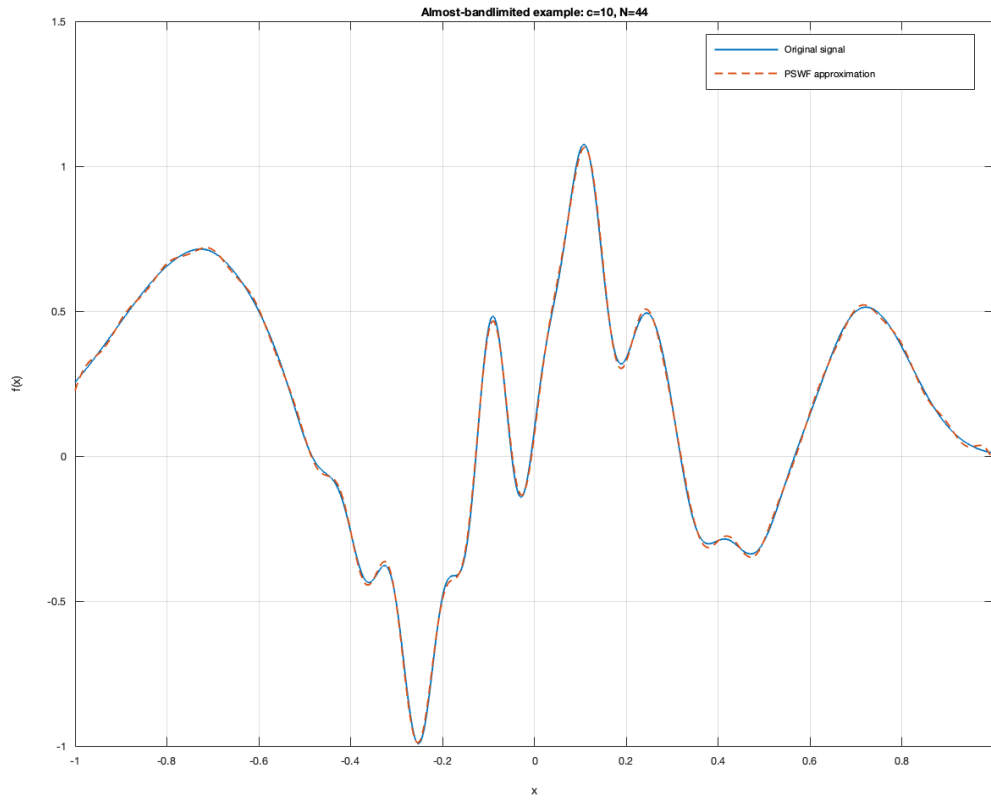


Figure 2: Almost-bandlimited test function and its approximation using the first 44 PSWFs with $c = 10$.

References

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